Stage 1 — Dataset Construction (12K Structured / Gold Decomposition Data)

MuSiQue (4K)

- Use original gold question decomposition
- Learn multi-hop chain decomposition

2Wiki (4K)

- Convert evidence triples to entity → relation steps
- Learn relation-style comparison decomposition

PopQA (4K)

- Convert KG triple to #1 : subject >>property
- Teach single-step is valid decomposition

12K Decomposition SFT Data

Total = 4K + 4K + 4K

Train decomposer to generate sub-questions, not answers

Stage 2 — Train Decomposer

12K decomposition data → LoRA fine-tuning → Decomposer model

Stage 3 — Inference (Dual-Path + Verifier)

Original Question

Direct Path: Original-Question RAG

Retrieve using original question → Answer with retrieved context → Record direct answer & confidence

Decomposition Path: Sub-Question RAG

(Sequential with Confidence-Based Retrieval & Answer Propagation)

1. Decomposer model
2. Generate sub-questions
3. Answer Sub-Q1 and measure confidence
   - high confidence → 5
   - low confidence → 4
4. If low confidence, retrieve & re-answer
5. Pass finalized Answer 1 into Sub-Q2
6. Repeat for next sub-questions (Sub-Q2 ... Sub-Qn)
7. Synthesize final answer

Verifier → Final Answer